Hybrid Rocket Thrust Curve Generator

Quick User Manual

Streamlit UI - current project configuration

Purpose
This manual explains how to launch the local Streamlit interface, change simulation inputs, run the existing hybrid-rocket model, review results, and generate an RSE file. The UI does not replace the underlying physics model.

1. Project Requirements

- Windows with PowerShell and Visual Studio Code (optional but recommended).
- Python 3.12 project virtual environment at .venv.
- MATLAB R2025b with MATLAB Engine for Python.
- MatlabCEA folder available beside the thrust-curve-generator folder and visible to MATLAB.
- Python packages installed from requirements.txt, including Streamlit, CoolProp, NumPy, pandas, SciPy, matplotlib, and matlabengine.

Recommended folder layout:

```
C:\Users\sota0\PROJECT\CSI\26-27\
|-- thrust-curve-generator\
|   |-- .venv\
|   |-- streamlit_app.py
|   |-- run_ui.ps1
|   |-- engine.py
|   |-- file_generator.py
|   `-- rse files\
`-- MatlabCEA\
```

2. Launching the UI

From VS Code

1. Open the thrust-curve-generator folder in VS Code.
2. Open Terminal > New Terminal.
3. Run the launch script:

```
.\run_ui.ps1
```

4. A browser window should open with the Hybrid Rocket Thrust Curve Generator interface, normally at http://localhost:8501.

Alternative PowerShell command

```
.\.venv\Scripts\python.exe -m streamlit run .\streamlit_app.py
```

First-time dependency update
If Streamlit or another package is missing, activate the project virtual environment and run: python -m pip install -r requirements.txt

3. Using the Simulation Inputs

The default values shown in the UI reproduce the current values used by main.py. The main input groups are:

UI Section	Inputs
Motor	Initial oxidizer mass, initial fuel mass, initial port diameter, initial oxidizer mass flow rate, nozzle expansion ratio
Pressure System	Chamber pressure, initial oxidizer tank pressure, pressurant temperature, initial COPV pressure, COPV volume
Vehicle / COPV	Minimum thrust setting, COPV length used by the mass model, COPV dry mass
Fixed Selections	Fuel = PARAFFIN, Oxidizer = LOX, Pressurant = Nitrogen
Advanced Mass Components	Mass, local CG, and length for chamber, nozzle, and interstage components

Current input checks: oxidizer tank pressure must exceed chamber pressure, and initial port diameter must stay below the current grain outer-diameter limit.

Important current-model behavior

Some UI inputs are coupled through derived design quantities. The current model can recalculate injector area, grain length, and oxidizer-tank length from selected inputs. Treat parameter sweeps as design changes unless the corresponding hardware is fixed in the underlying model.

4. Running a Simulation

1. Enter or adjust the desired inputs.
2. Click Run Simulation.
3. Wait while MATLAB/CEA and the Python model complete the calculation.
4. A success message appears when the simulation finishes.

No automatic file output

Run Simulation does not create an RSE file. This is intentional so that parameter studies can be run without generating unnecessary output files.

5. Reading the Results

The UI displays summary metrics including:

- Average thrust and peak thrust
- Total impulse and burn time
- Average specific impulse (ISP)
- Average O/F ratio
- Calculated grain length and oxidizer-tank length

Result tabs:

Tab	Contents
Thrust	Thrust history versus time
O/F	Oxidizer-to-fuel ratio history
Mass	Vehicle wet-mass history
CG	Center-of-gravity history

Tab	Contents
Detailed values	Initial/final propellant states, port diameter, flow rate, temperatures, nozzle diameters, thrust coefficient, mass fraction, and throat mass flux

6. Generating an RSE File

1. Run a simulation successfully first.
2. Scroll to the RSE Export section below the results.
3. Check the expected output path shown by the UI.
4. Click Generate RSE File.
5. The file is written into the project rse files folder using the existing file_generator.py logic.

RSE limitations

The current RSE generator supports the impulse-class ranges implemented in the existing code (M through S). If a file with the same generated motor code already exists, the UI warns that it will be overwritten.

7. Quick Troubleshooting

Issue	Action
run_ui.ps1 is not found	Confirm that run_ui.ps1 and streamlit_app.py are inside the thrust-curve-generator project folder.
Streamlit import error	Activate .venv and run: python -m pip install -r requirements.txt
MATLAB or CEA startup error	Run python .\check_environment.py and confirm that MATLAB Engine and CEA2 are detected.
CEA2 path issue	Run python .\setup_matlab_path.py and confirm that the sibling MatlabCEA folder is added to the MATLAB path.
Simulation input error	Check that oxidizer tank pressure is above chamber pressure and that the initial port diameter is below the grain outer-diameter limit.

8. Normal Workflow

Open project in VS Code -> run .\run_ui.ps1 -> edit inputs -> Run Simulation -> inspect results -> repeat as needed -> Generate RSE File only for the selected case

End of Quick User Manual